

Sealed-Bid auction scheme based on Blockchain and Secure Multi-Party Computation

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Abstract—With the rapid development of Internet technology, traditional offline auction is gradually transforming to electronic auction, in which the privacy of bidders is becoming more and more concerned. Most of the existing sealed electronic auction schemes guarantee the fairness of the auction through the third party auction house, but the third party's distrust also brings security risks to the privacy disclosure of the bidder's identity, price tag and so on. In view of the above problems, this paper proposes an auction scheme based on blockchain and secure multiparty computing. The scheme takes advantage of the characteristics of decentralized and tamper proof of blockchain to build an auction environment without third party. The idea of deposit on blockchain restricts the behavior of bidders and improves the security of sealed auction scheme. Secure multi-party computation and ring signature technology are used to protect the bidders' privacy from disclosure and public verifiability. Through security analysis, the proposed auction scheme meets the security requirements of electronic auction, and is more fair and reliable than the traditional auction scheme.

Keywords—electronic auction; blockchain; secure multi-party computing; ring signature; privacy protection

I. INTRODUCTION

With the rapid development of Internet technology and the rise of electronic commerce, electronic auction has become a way for people to auction goods. A good auction system can provide a fair trading environment for buyers and sellers, save users time and cost, and make people enjoy the convenience of technology without leaving home. However, while enjoying the convenience, the security problem of electronic auction has gradually emerged. Therefore, how to design secure electronic auction protocol has become one of the research hotspots in recent years.

Sealed auction can well protect the identity and price information of bidders other than the successful bidder, and protect the privacy of unsuccessful bidders. It is a more extensive form of auction in today's society. The scheme of this paper is about the first price sealed auction. In order to ensure that the seller does not collude with the bidder and the bidders do not collude with each other, the following security requirements should be met:

Fairness: Bidders have the same status to obtain information, and there is no one party can obtain more

information than the other party, which is more conducive to winning the auction.

Anonymity: The identity information of bidding participants must be kept confidential.

Secrecy of bidding price: After the completion of bidding, the bid prices of all bidders except the successful bidder shall be kept confidential.

Undeniability: The bidder can not deny his tender price after bidding.

Unforgeability: No one can forge the identity of the bidder to participate in the auction.

Public verifiability: Everyone can verify the identity of the successful bidder.

For sealed auction, many researches have been made by the predecessors. Franklin and Reiter[1] put forward a sealed security auction scheme for the first time. In this scheme, electronic money is used to protect the bid price from tampering and to restrain the bidder's behavior through a certain amount of money. Kikuchi et al.[2] proposed an auction scheme based on multi-party secret computing protocol. Xiao[4] analyzed the security defects of the scheme proposed by Yang[3]. On the basis of the original scheme, they promised a random number and a bid price by using the double commitment scheme, which made the improved scheme have the properties of unforgeability and non repudiation, but still can not meet the confidentiality of the bid price.

In the existing research on sealed auction, there are often third-party auction houses. However, the auction house will charge a lot of fees when carrying out the auction activities, which increases some extra cost. Brandt[5] designed an auction protocol that does not rely on a trusted third party to protect the privacy of bidders in the auction, but all the calculations in the protocol need to be completed by the seller himself. Cheng et al.[6] proposed a sealed electronic auction scheme based on the untrusted third party, which uses digital signature technology to verify the identity of bidders.

With the emergence of blockchain, people begin to gradually use blockchain technology to replace the third party. At present, some researches have solved practical problems through blockchain technology. Such as logistics, anti-counterfeiting traceability, electronic voting, electronic auction, etc. Peng[7] designed an electronic

auction protocol without the participation of a third party based on blockchain and concurrent signature. By using the fuzziness of concurrent signature, the identity privacy of bidders is protected. Li et al.[8] introduced a sealed bidding auction scheme based on blockchain smart contract technology, which constructs an auction environment without third party through blockchain.

In this paper, a sealed auction scheme based on blockchain and secure multi-party computing is proposed to overcome the shortcomings of the current auction scheme, such as the disclosure of identity information, the auction house may tamper with the winning price and so on. Bidders can sort the secret bid price through secure multi-party calculation. Combined with the idea of secure deposit on the blockchain, bidders are constrained to commit malicious acts to ensure that the winning bid price can not be tampered with, but also ensure the fairness of the auction.

II. PRELIMINARIES

A. Overview of blockchain

The concept of blockchain originated from an article “Bitcoin: A Peer-to-Peer E-cash System” published by Nakamoto in 2008 [9]. In essence, it is a decentralized distributed database, which adopts P2P communication mode, and has the characteristics of openness and non tampering. Due to its unique distributed characteristics, blockchain technology overcomes the dependence on servers and can be successfully applied to other industries except cryptocurrency and finance, such as medical treatment, intelligent transportation, microgrid, supply chain management, etc.

Bitcoin is the first successful application of blockchain technology. After bitcoin, other cryptocurrencies based on blockchain have been proposed, such as zerocoin and monero. The workflow of bitcoin is shown in Fig. 1.

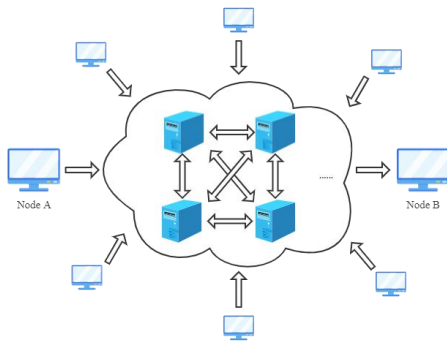


Fig. 1. Bitcoin workflow.

B. Secure multi-party computation

The concept of secure multi-party computing was first proposed by Professor Yao Qizhi in 1982, and a secure two-party comparison protocol was proposed to solve the

millionaire problem. A simple generalization of the millionaires problem is that multiple data are sorted by size without disclosing any information, which is obviously of great practical significance. In electronic auction, every bidder is not willing to disclose his bid price, but he needs to know the bidder with the largest bid price in this round of auction. The mathematical model of this problem is actually a generalization of the millionaire problem. In reference[10], Tang et al. constructed a secure multi-party sorting protocol for this mathematical model. By running the protocol, we can get the correct sorting of n secret data.

Initialization phase: Suppose the protocol has n participants, $P_1, P_2, \dots, P_n$, each participant P_i has an public identity x_i , and a secret number $s_i \in \mathcal{Q}^+$, $i = 1, 2, \dots, n$. There is a secret channel between any two participants, and each participant has a broadcast channel.

Input phase: For $i = 1, 2, \dots, n$ each P_i randomly and independently selects $2t+1$ t -degree polynomials to share $s_i, s_i^2, \dots, s_i^{2t}, s_i^{2t+1}$ in the rational number field,

$$\begin{aligned} f_{i_1}(x) &= s_i + a_{i_1 1}x + a_{i_1 2}x^2 + \dots + a_{i_1 t}x^t, \\ f_{i_2}(x) &= s_i^2 + a_{i_2 1}x + a_{i_2 2}x^2 + \dots + a_{i_2 t}x^t, \\ &\vdots \\ f_{i_{2t+1}}(x) &= s_i^{2t+1} + a_{i_{2t+1} 1}x + a_{i_{2t+1} 2}x^2 + \dots + a_{i_{2t+1} t}x^t \end{aligned} \quad (1)$$

Then P_i computes $f_{i_1}(x_j), f_{i_2}(x_j), \dots, f_{i_{2t+1}}(x_j)$ and sends it to P_j . In addition, the participant also generates a random number $r_i \in \mathcal{Q}^+$, and randomly picks a t -degree polynomial $r_i(x)$ in the rational number field, i.e., where (2). Finally, $r_i(x_j)$ is computed and send $r_i(x_j)$ to P_j , for all $j = 1, 2, \dots, n$.

$$r_i(x) = r_i + a_{i1}x + a_{i2}x^2 + \dots + a_{it}x^t \quad (2)$$

Calculation phase:

1) All participants agree to choose $2t+1$ random polynomials from $r_1(x), r_2(x), \dots, r_n(x)$. let the $2t+1$ random polynomials be $r_1(x), r_2(x), \dots, r_{2t+1}(x)$, which correspond to participants $P_1, P_2, \dots, P_{2t+1}$, respectively. Each participant P_j calculates $\lambda_i g_i(x_j)$, where (3).

$$g_i(x_j) = r_1(x_j)f_{i_1}(x_j) + \dots + r_{2t+1}(x_j)f_{i_{2t+1}}(x_j) \quad (3)$$

Each P_j in $P_1, P_2, \dots, P_{2t+1}$ randomly chooses n t -degree polynomials $h_{j1}(x), h_{j2}(x), \dots, h_{jn}(x)$, which are used to share $\lambda_j g_1(x_j), \lambda_j g_2(x_j), \dots, \lambda_j g_n(x_j)$, respectively, with $h_{ji}(0) = \lambda_j g_i(x_j)$, where $1 \leq i \leq n$.

Then P_j computes $h_{j1}(x_k), h_{j2}(x_k), \dots, h_{jn}(x_k)$ and sends them to P_k as shares, where $1 \leq k \leq n$.

Output phase: For each P_j , compute (6) and output, where $j = 1, 2, \dots, n$.

$$y_j = (y_{j1}, \dots, y_{jn}) = \left(\sum_{i=1}^{2t+1} h_{i1}(x_j), \dots, \sum_{i=1}^{2t+1} h_{in}(x_j) \right) \quad (6)$$

Implementation of sorting: Suppose each P_j has an output y_j ($1 \leq j \leq n$). Using the Lagrange interpolation formula, an arbitrary $t+1$ participants can jointly reconstruct (7), $1 \leq i \leq n$, and then broadcast (P_i, s'_i) at the same time. Therefore, anyone can use (P_i, s'_i) , to obtain the value sequence $s_1, s_2, \dots, s_n$.

$$s'_i = r_1 s_i + r_2 s_i^2 + \dots + r_{2t+1} s_i^{2t+1} \quad (7)$$

So far, the protocol is over. Through the analysis of the correctness and confidentiality of the protocol, it is proved that the protocol can get the correct order of data, and even if the participants have unlimited computing power, the confidentiality of the protocol still holds.

C. Ring signature

In 2001, Rivest proposed the concept of ring signature for the first time in an article entitled "How to Leak a Secret"[11], and presented an RSA version of ring signature scheme. The signature scheme consists of the following algorithms:

1) Sign(): A probabilistic algorithm generates a signature $\sigma = (m, L, c_i, x_1, \dots, x_r)$ for message m after inputting message m , a series of public keys $L = \{y_1, y_2, \dots, y_r\}$ and a private key S_s of ring members in the group.

2) Verify(): A deterministic algorithm, after inputting (m, σ) , outputs true if σ is the ring signature of m ; otherwise, outputs false.

III. BIDDING SCHEME

This section describes the basic process of the auction scheme. Blockchain secure deposit is used to restrict the bidders' behavior and force them to disclose the bid price

within a specified time, otherwise they will face the penalty of losing deposit.

A. Bidding Process

The auction scheme is divided into auction preparation stage, bidding stage, bid opening stage and auction completion stage. The following auction scheme is given. As shown in Fig. 2.

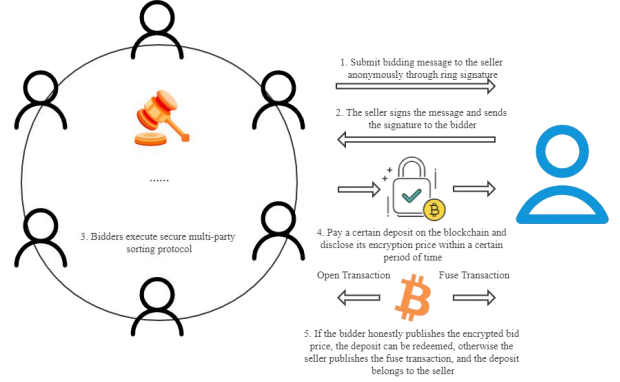


Fig. 2. Auction scheme.

All bidders under the chain through secure multi-party calculation, jointly calculate the size of the price order, select the winner of the highest bid price, the identity of the successful bidder exists in the form of $H(U_{id})$. At the end of bidding, the seller confirms the final price by opening the encrypted information on the chain. When the successful bidder and the winning price are determined, the successful bidder needs to provide U_{id} and r for all participants to verify. The overall auction process is shown in Fig. 3.

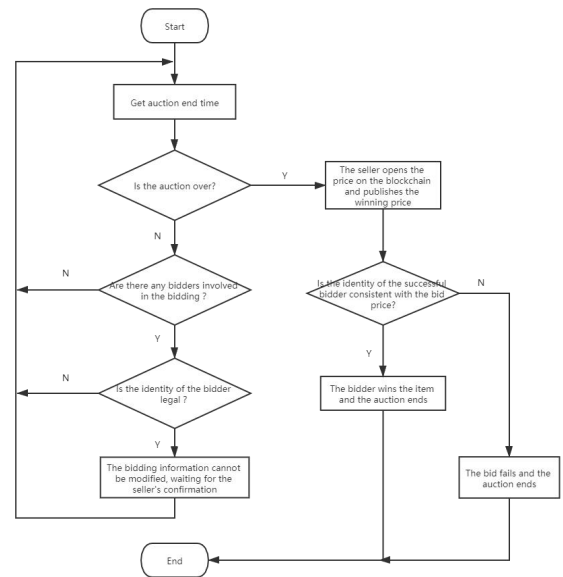


Fig. 3. Overall auction process.

B. Symbol description

TABLE I. SYMBOL DESCRIPTION.

Mark	Meaning
$Commit(in : T^{b_s})$	b_s uses T^{b_s} transaction as input to form <i>Commit</i> transaction
$Sig_{b_s}([Commit])$	b_s uses private key to sign <i>[Commit]</i>
<i>[Commit]</i>	All contents except input script in <i>Commit</i> transaction
<i>in-script</i> : $Sig_{b_s}([Commit])$	Input script parameter information
<i>out-script</i> (<i>body</i> , σ)	Output script parameter information
$Ver_{b_s}(body, \sigma)$	Use the public key of b_s to verify whether the digital signature σ is correct
$val : d \text{ BTC}$	The amount of the transaction is d BTC
$tlock : t$	The transaction is locked within time t , and can only be unlocked after time t

C. Auction preparation stage

Before the auction starts, the auction participants carry out the necessary initialization work, which is carried out on the blockchain. The seller publishes the detailed information of the auctioned goods, auction rules and deadline, etc. on the chain. Bidders are required to register with Registration Center, and the RC issues identity ID and system security parameters for bidders. All parties publish public key and bitcoin address to facilitate signature, verification of information and bitcoin transfer. There are n bidders, $b_1, b_2, \dots, b_n$ respectively. The corresponding public-private key pair is $((PK_1, SK_1), (PK_2, SK_2), \dots, (PK_n, SK_n))$. The seller is represented by P , and its corresponding public-private key pair is (PK_A, SK_A) .

D. Bidding stage

After the seller announces the start of the auction, the bidder starts to bid. Suppose that bidder b_s wants to bid at the price of m_s . The auction scheme is as follows:

Step1: Bidder b_s selects the bid price m_s and takes the hash value as the symmetric key key_s , where $key_s = H(m_s)$.

Step2: Pick a random value: b_s pick a random value v_s on $\{0, 1\}^*$.

Step3: Select $n-1$ random values: b_s selects the random value x_i on $\{0, 1\}^*$ for the group member b_i ($1 \leq i \leq n, i \neq s$), and calculates $y_i = g_i(x_i)$.

Step4: b_s to calculate the equation (8). The equation has only one solution y_s , which can be calculated.

$$C_{key_s, v_s}(y_1, y_2, \dots, y_n) = v_s \quad (8)$$

Step5: According to the trapdoor replacement of b_s , $x_s = g_s^{-1}(y_s)$ is calculated.

Step6: b_s calculates the hash value $H(U_{id} \parallel m_s \parallel r_s)$ of its bid commitment.

Step7: The bidder b_s submits the bidding message to the seller P in the form of (9).

$$(PK_1, PK_2, \dots, PK_n; v_s; H(U_{id} \parallel m_s \parallel r_s); x_1, x_2, \dots, x_n) \quad (9)$$

After receiving the bidding message, the seller signs the message and sends the signature to the bidder as a receipt. Bidders submit their own bid triplet (10),

$$(H(m_j), H(ID_j \parallel m_j \parallel r_j), v_j) \quad (10)$$

where v_j is convenient for sellers to verify, that is, they can correspond to the previous received bid messages one by one. At the same time, all bidders start to implement the secure multi-party sorting protocol with anonymous identity $H(U_{id})$.

E. Bid opening stage

At this stage, all bidders pay a deposit on the blockchain and disclose their encryption bid price within a certain period of time, otherwise the deposit belongs to the seller [12].

Still take the bidder b_s as an example, the process is shown in Fig. 4.

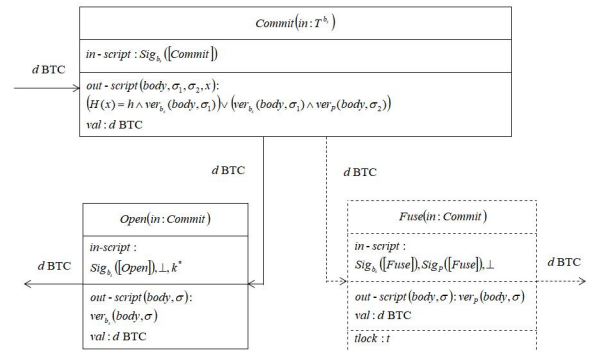


Fig. 4. Price opening.

Pre-condition:

Step1: The bidder b_s and seller P have their own bitcoin key (pk_s, sk_s) and (pk_A, sk_A) respectively, exchange their own public key and keep the private key.

Step2: The real price of b_s is k . After using the seller's public key encryption, it is k^* , and set (11).

$$h_{b_s} = H(k^*) \quad (11)$$

Step3: b_s specifies a transaction T^{b_s} with a value of d BTC as the deposit.

Commit phase:

Step1: b_s uses T^{b_s} as input to create a *Commit* transaction.

Step2: b_s creates a $Fuse^{b_s}$ transaction with a time lock and sends his signature to the seller together with the $Fuse^{b_s}$ transaction.

Step3: b_s signs the *Commit* transaction and sends the signature to the seller.

Step4: Seller P verifies the signature of bidder b_s . If the verification is successful, the private key sk_A is used to sign the *Commit* transaction and broadcast the whole network.

Step5: Both sides wait for the *Commit* transaction to be written into the blockchain.

Step6: If *Commit* transaction does not appear on the blockchain within the specified time, b_s immediately redeems transaction T^{b_s} and exits the protocol.

Open phase:

Step7: The bidder b_s broadcast $Open^{b_s}$ transaction and disclose the encrypted price k^* .

Step8: If the $Open^{b_s}$ transaction does not appear on the blockchain within t time, the seller P broadcast $Fuse^{b_s}$ transaction gets the d BTC deposit designated by the bidder.

F. Auction Completion Stage

After receiving the encrypted price k^* , seller P decrypts it with its private key to get the real price k , and then verifies it in the next stage according to the auction rules.

Step1: The seller compares all price sizes to get the maximum price k_i .

Step2: According to the trapdoor replacement, the corresponding $y_i = g_i(x_i)$ of x_i is calculated.

Step3: Find the corresponding bid triplet $(H(m_i), H(U_{id} \parallel m_i \parallel r_i), v_i)$ through v_i in the bid message, calculate the hash value of the bid price k_i , as is shown in $key_i = H(k_i)$, and verify whether $key_s = key_i$ is true.

Step4: Test whether the values of $y_1, y_2, \dots, y_n$ satisfy the equation of ring equation (12). If the equation is true, the identity of the bidder is proved to be legal. Otherwise, the seller judges that the bidder's identity is illegal and does not belong to the ring signature group and excludes it.

$$C_{key_i, v_i}(y_1, y_2, \dots, y_n) = v_i \quad (12)$$

After the bidders execute the secure multi-party sorting protocol, each party can know the ranking order of the bid price, but not the further information (the real bid price). Then the winner is selected according to the auction rules, and the winner provides his own U_{id} for all participants to verify, that is, $H(U_{id}) = true$. The seller publishes the signature message of the successful bidder on the blockchain, bid commitment $H(U_{id} \parallel m \parallel r)$, bid triplet $(H(m), H(U_{id} \parallel m \parallel r), v)$, real bid price k and encrypted bid price k^* . The successful bidder provides his own random value r . Anyone can verify the identity of the successful bidder and the correctness of the winning price according to U_{id} and r .

IV. SECURITY ANALYSIS**A. New protocol security analysis**

In this paper, a new sealed electronic auction scheme is designed by combining secure multiparty computation, ring signature and deposit on blockchain.

Correctness: The bidder selects the successful bidder by implementing the secure multi-party sorting protocol, and only when the bidder honestly publishes his encryption bid price on the blockchain, the deposit can be redeemed, that is, releasing the correct k^* to make the equation $H(k^*) = h$ hold. Then the seller uses the private key to decrypt the real price k , and selects the winner according to the auction rules.

Fairness: Users on the blockchain have equal status. Due to the openness and transparency of data, all users get the same information, and there is no case of who has privileges. If the bidder is honest, the bidding message can

be submitted successfully under the chain, and the encrypted bid price will be made public within the specified time on the chain, so that the auction can proceed smoothly.

Anonymity: In the bidding stage, because of the anonymity of the ring signature itself, even the seller can not know the specific identity of the bidder. By verifying whether the user's identity is legal, it can determine whether the bidder belongs to the group.

Secrecy of bidding price: In the bidding stage under the chain, the unidirectionality of hash function makes it impossible for anyone to restore the real bid price even if they get the bid message and bid triplet. In the open stage of the chain, the bidder releases the encryption price k^* , and only the seller can decrypt it with his private key.

Undeniability: Due to the non tamperability of the blockchain, the successful bidder can not deny his bid price.

Unforgeability: Only members in the ring can bid, and no one can forge the identity of members in the ring, which has certain exclusivity.

Public verifiability: All participants can verify the identity and bid price of the successful bidder according to the signature message disclosed by the seller on the blockchain, the bid commitment $H(U_{id} \parallel m \parallel r)$, the bid triplet $(H(m), H(U_{id} \parallel m \parallel r), v)$, the true bid price k , the U_{id} and random value r provided by the winning bidder.

B. Scheme comparison

F. Brandt combines the knowledge of matrix with distributed El Gamal public key cryptography constitution, and successfully cancels the third party organization, and applies secure multi-party computing to the electronic auction scheme. Xiao proposes an auction scheme based on hash chain, which uses double commitment to achieve non repudiation, but there are still third-party auction houses. In this paper, the combination of blockchain and secure multiparty computing, compared with F. Brand's scheme and Xiao's scheme, has higher security performance. The specific comparison is shown in Table 2.

TABLE II. SCHEME COMPARISON.

	F. Brandt's scheme	Xiao's scheme	The scheme of this paper
No third party	√	×	√
Open and transparent	×	√	√
Price confidentiality	√	×	√

Non repudiation	√	√	√
Computationally efficient	×	√	√
Anti conspiracy attack	×	√	√

V. CONCLUSION

In this paper we proposes a sealed auction scheme based on blockchain and secure multiparty computing. The idea of deposit on blockchain is used to restrict the behavior of bidders, and the extra cost of the third-party auction house is saved. Combined with secure multi-party computing and ring signature technology, the privacy of bidders is protected while the fairness of auction activities is ensured. Through the security analysis of the scheme, the scheme meets the basic security requirements of the electronic auction scheme, and achieves the correctness, fairness, anonymity, price confidentiality, non repudiation, non forgery and public verifiability of the successful bidder required by the sealed auction scheme.

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